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Course: Calculus Beta

Assignment: MyLab Math - Uge 10

Solve the differential equation.

$$9y' = 4e^{x/9} + y$$

To solve the differential equation, put the given equation into the standard form for a first-order linear differential equation. After the equation is in this form, it may be solved using an integrating factor.

Write the given equation so that it is in the standard form $y' + P(x)y = Q(x)$.

$$9y' = 4e^{x/9} + y$$

$$9y' - y = 4e^{x/9} \quad \text{Subtract } y \text{ from both sides of the equation.}$$

$$y' - \frac{y}{9} = \frac{4}{9}e^{x/9} \quad \text{Divide both sides of the equation by 9.}$$

Find the integrating factor $v(x) = e^{\int P(x)dx}$. First identify $P(x)$ from the general form equation.

$$P(x) = -\frac{1}{9}$$

Find the indefinite integral.

$$\begin{aligned} \int P(x)dx &= -\frac{1}{9} \int dx \\ &= -\frac{1}{9}x + C \quad \text{Integrate.} \end{aligned}$$

Take the simplest antiderivative, $C = 0$, and simplify the resulting expression for $v(x)$.

$$\begin{aligned} v(x) &= e^{\int P(x)dx} \\ &= e^{-x/9} \quad \text{Substitute } \int P(x)dx = -\frac{x}{9}. \end{aligned}$$

Now multiply both sides of the equation by the integrating factor, simplify, and integrate both sides of the equation..

$$\begin{aligned} y' - \frac{y}{9} &= \frac{4}{9}e^{x/9} \\ e^{-x/9}y' - \frac{1}{9}e^{-x/9}y &= \frac{4}{9} \quad \text{Multiply both sides of the equation by } e^{-x/9}. \\ \frac{d}{dx}(e^{-x/9}y) &= \frac{4}{9} \quad \text{Product rule.} \\ \int \frac{d}{dx}(e^{-x/9}y) dx &= \frac{4}{9} \int dx \quad \text{Integrate both sides of the equation.} \\ e^{-x/9}y &= \frac{4}{9}x + C \quad \text{Simplify.} \end{aligned}$$

Finally, solve the resulting equation for y .

$$y = \frac{4}{9}xe^{x/9} + Ce^{x/9}$$